

Structures of Reality and Simulation: A Reference-Frame Study of Root-Level, Reality-Level, Believed-Real, and Simulation Structures

Research Foundation, Project Assessment, and Team Roadmap — OAL-STRUCT-001-CLAUDE · v1.0 · Draft — Under Review

Table of Contents

1. Document Information

Field	Value
Title	Structures of Reality and Simulation: A Reference-Frame Study of Root-Level, Reality-Level, Believed-Real, and Simulation Structures
Document ID	OAL-STRUCT-001-CLAUDE
Version	1.0 (draft)
Date	September 23, 2026
Prepared by	Claude (Anthropic) — Primary Audit and Documentation Layer
Architect of record	Nathan Lewis II
Series and track	Foundation track (OAL-STRUCT). It supplies the reference frame. The earlier documents (OAL-CFTRL-RS-001, -LAB-001, -PLAN-001) belong to the application track (Conditional Failure at the Root Level, CFTRL), which this document is meant to inform.
Why the title changed	The Architect noted that nearly every title in the team’s responses, including those from this layer, centered on Conditional Failure at the Root Level. The intended subject is the structures themselves. CFTRL is what the knowledge is later used to design.
Research method	Web research on September 23, 2026, preferring primary and near-primary sources. Each source’s verification status is in Section 14.
Canon status	DRAFT — UNDER REVIEW. ONESTAMP: PENDING (Architect). Nothing here is canon.

Status tags used in this document. [V] verified against a source retrieved in this session. [P] partly verified (abstract, reference entry, or secondary description). [N] not re-verified in this session. [A] analysis by this layer, offered for review. [H] hypothesis to be tested.

2. My Understanding of the Correction

Yes, I understand. Restated in my own words so the Architect can correct it:

1. The subject of this study is **structures**: root-level structures, reality structures, structures that people believe are real or literal, simulation structures, and the frameworks and designs of each.
2. The reason is that a project cannot design around root-level failure without a **reference frame**: a map of what is known, believed, and modeled about what things are and how they are built. The map may be incomplete, and may be built in several competing ways.
3. The reference frame is **not an assertion** that we are, or are not, in a simulation, or that we know what is real, what is not, what base reality is, or what a reality-level structure is. It says: *we have studied this information, and here is what it says*.
4. The questions to work through are: What are root-level structures, and what do people believe they are? What is considered real, not real, and simulated? Can reality be distinguished from a simulation, and how would we know? How do we build a map of reality, and a map of a simulation? Do the two share the same structure?
5. Only after that is the knowledge **used** to inform the design and development of CFTRL.

Everything below is organized to follow that order. CFTRL appears in Section 11 as an application, not as the title or the frame.

3. How This Topic Arose

The concept began on March 22, 2026, when the Architect dictated **Conditional Failure at the Root Level** and **the Department of Endings**, documented in parallel by Perplexity and ChatGPT. On September 23, 2026, reviewing the Master Change Log and Master Development Log, the Architect saw that many concepts hold a name and a skeleton but lack supporting detail. Reflecting on CFTRL produced questions about what root-level structures are, what is real, whether reality can be told from a simulation, and whether a reliable map can be built. The AI collaborators responded, mostly in terms of CFTRL. The Architect then clarified that the aim is to move from thinking about CFTRL to studying the underlying structures first.

Date	Event
March 22, 2026	Architect’s voice dumps; CFTRL and the Department of Endings documented (Perplexity, ChatGPT)
Sep 23, 2026	Request 1 and Request 2: research on root-level structures, base reality, failure detection; then how to study what lies beyond, laboratory design, reliable maps, failure as an example, and non-assumptions
Sep 23, 2026	Request 3: documentation must state title, date, origin, verbatim prompt, purpose, goal, assessment, direction, and team plan
Sep 23, 2026	Request 4 (this correction) : shift the subject from CFTRL to root-level, reality, believed-real, and simulation structures and their frameworks, as a reference frame that makes no assertion about what is real

4. Verbatim Origin Prompts

The Architect's words are reproduced without change. Each original line is a separate paragraph. Phrases in the non-assumptions list are conditions on the design and must be quoted together with their qualifying clauses.

4.1 Scope correction (September 23, 2026) — the most recent statement

Okay, so I was reading through the responses and I noticed there is a discrepancy, Everyone pretty much talked about conditional failure at the root level, which is fine, but what I actually asked for was the following: I noticed in every single title that it was almost with all of them conditional failure at the root level. The goal is to transition from thinking about conditional failure at the root level to:

- Root level structures
- Reality structures
- Structures that people think are real, literal
- Simulation structures
- Their framework, their design, things like that The reasoning behind that is that if you do not have a map of reality as you know it—and this is not to say that I understand reality or that I know what reality or what being real is—but it is just to say that we have to have a reference frame. We have to be able to say we have studied this information. This is not an assertion that we understand in whole or in part that either we are in a simulation or we are not in a simulation. We understand what is real, what is not, what is base reality, what is a reality level structure. We are trying to learn more about root level structures, reality structures, and what we understand now so that we can use that information to help aid in the design and development of conditional failure at the root level. But we have to learn about root level structures, what we believe they are, what is considered real, what is considered not real, and what is considered a simulation. Can we distinguish what is reality from a simulation? How do we know? How do we build a map of reality? How do we build a map of a simulation? Do those architectures share the same structure? We have to ask all of these questions and then use the information towards what we are trying to design. That's just the basis that I have starting out. Do you understand what I'm saying?

4.2 Initial statement (September 23, 2026)

Can you save this thought for me?

I need you to gather research and analysis for a concept.

Conditional failure at the root level.

The study of root level structures, how they arise, how they fail, how they are maintained. Please include the original documents for this content that were generated.

How do you know the difference between a root level structure and a structure that is not? How can a system distinguish between base reality and something that is not? How can a system determine when a root level failure has occurred or when base reality has been compromised.

How do we study this? What about this concept works, what doesn't? What questions do we need to ask?

4.3 Expanded statement (September 23, 2026)

How can we study what lies beyond these structures? Can we model whether structures in reality follow a similar design? Or diverges in ways we haven't thought of? Is this feasible using simulations or another type of modeling? How? Can we design a laboratory environment to do this? How? Can you build a reliable map of reality? What basis of comparison do you have when you are operating on incomplete information? Use failure as an example to learn from.

What I do not assume in the design of this:

That I have the complete architecture and map of all simulations, of reality, of existence as we know it.

That we share a unified definition of what reality is.

That existence can be fully mapped, represented, comprehended in full, or that this is the aim of this laboratory.

That I aim to cause a root level compromise of simulations, of base reality, or any other structure.

That this is a how to guide to do any of those things.

It is intended to help in the design of simulation and reality level structure lifecycle design, maintenance, evolution, and improvement.

Within this design frame It does not assume that we have complete knowledge or understanding of what that is.

How This topic arose from me thinking about conditional failure at the root level The master change log and master development log. I realized that there are a lot of concepts within that framework that have a definition or a name and a skeleton attached to them. They are a rough draft, they are an incomplete outline, there are a lot of additional details and supporting information that need to be added. The goal here is to design documentation to help support, design, and develop this concept. What I will include are the responses of all of the other AI collaboration team members, where they have generated documentation, I will include the documentation, and then I will include this complete response as well.

4.4 Clarifying instruction on documentation (September 23, 2026)

My task for you is to do research to support an outline of this design, development, and framework. If that was not clarified in the beginning i am clarifying now. the documentation that you generate should have a proper title, date, how this topic arose, the verbatim origin prompt from me, purpose, and goal. it should create an

assesment of where we are with this project and where we are trying to go and how you plan on helping the entire team get there. please be detailed in your research and analysis. please generate in pdf and docx. I am providing this prompt to you because some of the other ai's needed additional clarification.

5. Purpose

1. To build the **reference frame** the Architect described: a documented, sourced map of what is known, believed, and modeled about root-level structures, reality structures, believed-real structures, and simulation structures, and of the frameworks and designs each uses.
2. To do so **without asserting** what is real, what is simulated, or what base reality is.
3. To answer, as far as the evidence allows, the questions the Architect set: whether reality can be distinguished from a simulation and how we would know; how to build a map of reality and a map of a simulation; and whether their architectures share a structure.
4. To assess where the project stands under this corrected scope, state where it is going, and describe how this layer will help the whole team.
5. To pass what is learned to the CFTRL design work as an application.

6. Goal

Overall goal. A set of documents and a working method that let the team say, of any claim about root-level, reality, believed-real, or simulation structures: *this is what the sources say, this is who holds the view, this is the evidence, and this is how it compares with the others.* The knowledge then supports the design, development, maintenance, evolution, and improvement of structure lifecycles, including CFTRL.

#	Working goal	Evidence it has been reached
S1	A shared statement of the reference-frame principle and vocabulary	Approved by the Architect; cited by collaborators
S2	A Belief Atlas: families of views about what is real, with sources, holders, and evidence status	Each family has sourced entries; none is endorsed or dismissed
S3	A Root Atlas: root-level structures across domains, with how they arise, are maintained, and fail or change	Each entry has a sense of “root,” a source, and a status tag
S4	A Simulation-Structure Atlas: how simulations are built and what they imply	Engineered and hypothesized simulations kept separate
S5	A Structure Comparison Matrix across natural, engineered, social, and hypothesized-simulated structures	Approved dimension list; cells tagged by evidence
S6	A statement of what can and cannot be	Distinguishability table adopted

#	Working goal	Evidence it has been reached
	distinguished, and the evidence classes for each	
S7	A layered map schema for reality and for simulations	Schema adopted for new claims
S8	A bridge from the frame to the CFTRL design	CFTRL documents cite the frame

7. The Reference-Frame Principle

The Architect’s position, that a reference frame is needed without an assertion about what is real, has a well-developed counterpart in philosophy, and using it clarifies what the team is and is not doing.

Carnap’s internal and external questions [V]. In “Empiricism, Semantics, and Ontology” (*Revue internationale de philosophie*, 1950, vol. 4, pp. 20–40), Rudolf Carnap distinguishes two kinds of existence questions. *Internal* questions ask whether something exists within a chosen framework, and can be answered by the framework’s logical or empirical methods. *External* questions ask about the reality of the framework itself. Carnap holds that the external question is really a practical one: whether to adopt the framework, judged by how expedient or fruitful it is, and not a factual question that can be true or false. His view has critics: Quine’s objections are well known, and later philosophers have argued that the distinction is problematic. It is offered here as one supporting position, not as the foundation.

What this gives the team [A].

- **State the frame.** Every claim is made *within a stated framework* (a scientific model, a legal system, a simulation design, a belief tradition). The project then answers internal questions by that framework’s own rules.
- **Treat the choice of framework as a separate, practical question.** Which frameworks the team studies, and how they compare, is a decision about usefulness. It is not an assertion that any framework describes what is ultimately real.
- **Keep external claims out of the record as facts.** Claims about what is ultimately real, or whether we are simulated, are recorded as claims held by named sources, with their evidence tier.

Eight principles for the study [A].

#	Principle
P1	A reference frame, not an assertion. No document in this track asserts what is real or whether we are simulated.
P2	Beliefs about reality are studied as objects: who holds them, on what basis, with what structure. Recording a view is not endorsing it.
P3	Four kinds of statement are kept apart: what is <i>claimed</i> , what is <i>observed</i> , what is <i>modeled</i> , and what is <i>assumed</i> .

#	Principle
P4	Frameworks and designs are studied for their structure: levels, rules, state, time, resources, boundaries, observers, foundations, and endings.
P5	Architectures are compared through structure-preserving mappings, not by asserting identity.
P6	Root-level structures are studied in every domain before conclusions are drawn for CFTRL.
P7	Each claim carries a status tag and a source.
P8	Completeness is not the aim. Unknown regions are recorded as entries.

8. Belief Atlas: Structures That People Take to Be Real

This section maps the main families of views about what is real. It is descriptive. Each row states what the family holds, what counts as real within it, where its foundation lies, and the status of the entry. It does not rank the families.

Family	What it holds	What counts as “real”	Foundation (root)	Status
Scientific realism and structural realism	Mature science describes the world. Worrall’s epistemic structural realism holds that our scientific knowledge concerns structure. Ladyman and Ross’s ontic version holds that the world itself is structure.	What our best theories posit, or the structure they describe	Fundamental laws or structures; disputed which	[V] (Stanford Encyclopedia of Philosophy; Ladyman and Ross 2007)
Empiricism and non-realist structuralism	Van Fraassen defends an empiricist, non-realist structuralism about science. Theories are tools for organizing what is observable.	The observable; theory need not be believed	The observations themselves	[P] (named in the Stanford entry)
Framework relativism (Carnap)	Existence claims are relative to a chosen framework; the choice is practical.	An element of the framework	The decision to adopt the framework	[V]
Computational and informational views	Physical systems register and process information (Lloyd); the universe may be regarded as performing	Information and its processing	Physical law and the information it permits	[V] for Lloyd; [N] for Wheeler, Zuse, Fredkin,

Family	What it holds	What counts as “real”	Foundation (root)	Status
	a computation.			Wolfram, Tegmark
Simulation realism	Virtual worlds are genuine realities; virtual objects are real, differently constituted (Chalmers, <i>Reality+</i> , 2022).	Whatever has causal powers and is mind-independent and non-illusory	Whatever hosts the simulation	[V]
Simulation argument	At least one of three propositions holds, one being that we are almost certainly living in a computer simulation (Bostrom, 2003).	Not a theory of the real; an argument about probabilities	The hypothetical hosting civilization	[V]
Institutional and social reality	Facts such as money and nations depend on collective acceptance and assigned status functions. “X counts as Y in context C” (Searle, 1995).	Brute facts exist independently; institutional facts exist by acceptance	Collective intentionality	[V]
Legal and normative reality	A legal system’s validity rests on a foundation that cannot be justified by the system itself: Kelsen’s presupposed basic norm; Hart’s rule of recognition, grounded in official acceptance.	Validity within the system	The basic norm, or the rule of recognition	[P] (secondary sources)
Mathematical and abstract realism	Abstract entities such as numbers may or may not exist independently. Carnap treats the question as internal or pragmatic.	Varies by view	Axioms and definitions	[P] (Carnap); [N] (Platonism)
Idealist and phenomenalist views	Reality is mind-dependent or known only as it appears (for example, Berkeley; Kant’s distinction between appearance	What appears to minds	Mind, or the unknowable thing in itself	[N]

Family	What it holds	What counts as “real”	Foundation (root)	Status
	and thing in itself)			
Religious, spiritual, and cultural cosmologies	Many traditions describe layered or dependent realities and ask whether ordinary experience is ultimate. Chalmers cites the Zhuangzi butterfly dream, Narada’s transformation, and Plato’s cave as historical forms of the question.	Varies by tradition	Varies by tradition	[P] for Chalmers’s examples; the traditions themselves are on the research queue (Section 12)
Perceptual and cognitive models	Perception may be a model the brain builds and updates	What the model represents	The organism’s predictive machinery	[N]

Observations from the Belief Atlas [A].

1. **The families disagree about what the foundation is, not about whether there is one.** Nearly every entry has a foundation: fundamental structure, hosted computation, collective acceptance, a presupposed norm, or an adopted framework. That is the point at which “root-level structure” becomes a cross-domain idea.
2. **The foundation is often accepted, not derived.** In the legal, social, mathematical, and framework families, the foundation is presupposed or adopted. The Agrippan trilemma (Section 9) explains why.
3. **“Real” has at least four working meanings:** independent of minds (brute), dependent on acceptance (institutional), an element of a framework (Carnap), and having causal powers as a hosted object (Chalmers). A statement that something is “real” is incomplete without saying which.
4. **The historical form of the question is old and cross-cultural,** which supports treating it as a stable subject of study, not a passing idea.

9. Root Atlas: Root-Level Structures Across Domains

A root-level structure is treated here in the four senses used in earlier documents: **dependency** (what everything else relies on), **authority** (what sets the rules), **trust** (where verification stops), and **ontological** (what is not itself produced by something else), with **epistemic** (where justification stops) added. The table gives one entry per domain.

Domain	Candidate root (sense of root)	How it arises and is maintained	How it fails or changes	Status
Computer security	Root of trust (NIST: highly reliable components performing critical security functions; “a starting point that is implicitly trusted”) — <i>Trust</i>	Arises: by design. Maintained: secure design; verification by independent means (for example, Wheeler’s diverse double-compiling).	Subversion, as in Thompson’s compiler attack, which can leave source code clean	[V]
Measurement	The SI defining constants (Nov 2018 vote; in force May 20, 2019): kilogram, ampere, kelvin, and mole defined by fixing values of h, e, k, and N _A ; the platinum-iridium prototype kilogram retired as the definition — <i>Authority, reference</i>	Arises: by agreement of the member states of the Metre Convention. Maintained: experiments that realize the units; guarantee of continuity.	Replaced without changing the value of any unit. A root migration	[V]
Law	Kelsen’s basic norm; Hart’s rule of recognition — <i>Authority</i>	Arises: presupposed (Kelsen); a social practice of official acceptance (Hart). Maintained: continued acceptance by officials.	Loss of acceptance; sources say the ultimate criteria cannot be justified by the law itself	[P]
Epistemology	The foundation of a chain of justification: regress, circle, or accepted starting point (Agrippa’s trilemma, called the Münchhausen trilemma by Hans Albert) — <i>Epistemic</i>	Arises: every justification chain ends in one of the three. Maintained: by choosing which horn to accept: foundationalism, coherentism, or infinitism.	Discovery that the starting point was wrong	[V]
Social institutions	Collective acceptance and assigned status functions (Searle) — <i>Authority, dependency</i>	Arises: through shared attitudes, usually expressed in language. Maintained: continued shared acceptance and practice.	By the account, withdrawal of acceptance ends the institution [A]	[V] account; [A] failure reading
Network	Backbone network and	Arises: by design	A maintenance	[V]

Domain	Candidate root (sense of root)	How it arises and is maintained	How it fails or changes	Status
infrastructure	DNS (the 2021 Facebook outage) — <i>Dependency</i>	and accretion. Maintained: maintenance, audit tooling, out-of-band access.	command, an audit-tool bug, and a protective rule combined to remove the network and the tools needed to repair it	(company post-mortem)
Simulation engineering	The simulator and its host, with the experimental frame that defines what the model is stimulated by (Zeigler's DEVS framework, 1976) — <i>Dependency, authority</i>	Arises: by design. Maintained: by verification and validation ("living with error").	Divergence between model and system; resource exhaustion	[V]
Physical cosmology	Fundamental laws and initial conditions; the finite computational and information capacity of the universe (Lloyd: no more than 10^{120} operations on 10^{90} bits) — <i>Ontological (candidate)</i>	Arises: not settled. Maintained: not applicable.	Not settled	[V] for Lloyd's bound; the rest [H]
Biology	The genetic code and the last common ancestor of life — <i>Dependency, historical</i>	Arises: by evolution. Maintained: by heredity and repair.	Extinction or loss of a lineage	[N]
Mathematics	Axioms; limits on self-proof (Gödel) — <i>Epistemic</i>	Arises: by choice. Maintained: by consistency checking.	Discovery of inconsistency	[N]

Cross-domain pattern [A]. Across these entries a root falls into one of four kinds:

1. **Presupposed or accepted** (trust roots, basic norms, axioms, framework choices). The root cannot be justified from within. Its failure is often a *loss of acceptance or trust*, not a loss of function.
2. **Fixed by convention against invariants** (the SI). The root is deliberately tied to quantities chosen for stability. The SI shows that a root can be *migrated* without changing what depends on it, when continuity is designed in.
3. **Built and maintained** (infrastructure, simulators). The root is a maintained artifact. Its failure is a *maintenance, verification, or coupling* failure.

4. **Unexplained or given** (fundamental laws, initial conditions). Nothing is known about its origin or maintenance. Claims here are hypotheses.

Three consequences for the study [A].

- **The Agrippan trilemma [V] explains why roots are accepted.** Any justification ends in a regress, a circle, or an accepted starting point. A root is where that stopping point sits. This is a structural reason roots are unverifiable from below, and it holds in domains that have nothing to do with computers.
- **Loss of acceptance is a distinct kind of root failure.** Technical root-failure vocabulary (halt, corrupt, starve) does not describe a legal or social root that fails when officials or participants stop accepting it. CFTRL will need a class for this (Section 11).
- **Root migration is a designable pattern.** The 2019 SI change is a documented case of replacing a physical-artifact root with a set of fixed constants while guaranteeing continuity. It is a real precedent for the question “can a system move from one root to another without ceasing to be the same system?” that ChatGPT raised.

10. Simulation Structures

10.1 Engineered simulations: the design is known

Framework [V]. Zeigler’s *Theory of Modeling and Simulation* (1976; later editions 2000 and 2018) separates a **model** (the responding object), an **experimental frame** (the stimulating environment), and a **simulator** (the event driver) that interact by defined rules, and treats **system morphisms** (abstraction, representation, approximation) as the way models relate to the systems they stand for. The book’s own chapter titles include verification and validation under the heading “Living with Error.”

Why this matters here [A].

- It is a worked-out *framework and design* for simulation structures, which is what the Architect asked to study.
- The **experimental frame** is what determines what an observer of a model can see. That is the same idea as the “observation port” in LAB-001. It means “what can an analyst inside know” is a design choice in every simulation, and a study of it does not depend on any claim about reality.
- The **morphism** idea is a precise way to ask whether two architectures “share the same structure”: ask whether there is a structure-preserving mapping between them, and at what level of detail.

10.2 Hypothesized simulations of reality: the design is unknown

Bostrom’s argument [V] (*Philosophical Quarterly* 53(211), 2003, pp. 243–255) states that at least one of three propositions is true: the human species is very likely to go extinct before a posthuman stage; any posthuman civilization is extremely unlikely to run a significant number of simulations of its evolutionary history; or we are almost certainly living in a computer simulation.

Chalmers’s position [V] (*Reality+*, 2022) holds that virtual reality is genuine reality, that virtual objects are as real as non-virtual ones though differently constituted, and that we cannot know whether we are in a simulation.

Resource limits [V]. Lloyd’s calculation (*Physical Review Letters* 88:237901, 2002) gives the universe’s maximum capacity as about 10^{120} elementary operations on 10^{90} bits, and notes that the numbers are lower bounds on what simulating the entire universe on a quantum computer would need.

An implication to test [A][H]. If a simulated layer had to reproduce all of its host’s detail, it would need at least the host’s resources. So nested layers would in general be limited to reduced detail, smaller extent, or lower speed than their hosts. This is an implication of resource limits, not an established finding, and it could be studied in built stacks. It bears on Question 13 of RS-001, on whether depth inhibits diagnosis.

A tested example of a signature [V]. Beane, Davoudi, and Savage (arXiv:1210.1847, 2012) explored observable consequences of the hypothesis that the universe is a numerical simulation on a cubic space-time lattice. They assumed an early simulation with unimproved Wilson fermion discretization, and derived a bound on the inverse lattice spacing from the cosmic-ray energy cut-off. They proposed that such a simulation could show rotational symmetry breaking in the highest-energy cosmic rays. Commentary notes two caveats: the lattice might be built differently, and the effect would be measurable only if the lattice cut-off matched the known cut-off.

What it shows [A]. A test of this kind is **specific to an assumed simulation design**. A null result rules out that design, not “being simulated” in general. This is the essential difference between the two cases in Sections 10.1 and 10.2: for an engineered simulation the design is known and can be mapped; for a hypothesized one, each candidate design is a separate hypothesis with its own tests.

10.3 Contested formal arguments [V]

A 2025 paper by Faizal, Krauss, Shabir, and Marino (arXiv:2507.22950) argues from Gödel, Tarski, and Chaitin that a fully algorithmic theory of everything is impossible and that the universe therefore cannot be a simulation. Published discussion and online replies dispute the last step, arguing that limits on proving statements differ from limits on running a process. The entry is recorded as **contested**. The framework does not depend on it.

11. Can Reality Be Distinguished from a Simulation? How Would We Know?

The question has four layers, and different answers apply to each.

Layer	Answer	Basis
Within a framework	Yes, by the framework’s own rules. Something is “real” in the Carnapian sense when it fits the framework’s rules and evidence.	[V] Carnap

Layer	Answer	Basis
Against a specific simulation design	Sometimes. A design that predicts observable signatures can be tested, and a null result rules out that design.	[V] Beane et al.
Against any simulation whatever	Not in general. If the host controls everything the observer can perceive and measure, no internal observation need separate the cases. Chalmers holds that we cannot know.	[V] Chalmers; [A] for the general limit
By formal proof that reality is or is not computational	Contested.	[V] that the paper and replies exist

11.1 Distinguishability table

Hypothesis class	Kind of test	What a null result rules out	What a positive result would suggest	Status
Lattice-type simulation	Signatures of discretization (for example, cosmic-ray anisotropy)	That specific lattice design	A lattice-like structure	[V]
Resource-limited simulation	Fidelity, speed, or extent limits [H]	Little; the test is not yet defined	A resource-bounded host	[H]
Externally modified simulation	Unexplained rule or state changes	Little; an absence of interventions proves little	Outside interference or an internal cause not yet known	[A]
Perfect simulation	None available from inside	Nothing	Nothing	[A]
Non-algorithmic reality	Formal arguments	Not applicable	Contested	[V] contested

11.2 How do we know? [A]

We know only **relative to a framework and an evidence class**. For each claim the study records: the framework (what does “real” mean here?), the evidence (observation, model, formal argument, or assumption), the tier (empirical, testable in a lab, formal, or not resolvable), and what result would count against it. The Architect’s non-assumption that the

team does not presume a shared definition of reality is therefore built into the method, not just stated.

12. Building a Map of Reality and a Map of a Simulation

12.1 What a map is [A]

A map is a **structure-preserving representation**. Zeigler's system morphisms formalize the idea for simulations, and structural realism, in both its epistemic and ontic versions, is a view about what such representations capture: structure. Korzybski's line that the map is not the territory and Box's remark that all models are wrong but some are useful [N] point the same way. The map cannot be checked against the territory directly. It is checked against observation, against other maps, and by its predictions.

12.2 A layered map of reality (as known)

Layer	Content	Example	Typical evidence
L0 Observation	Recorded measurements	Cosmic-ray spectra; incident records	Data
L1 Reference frames	How quantities are defined and shared	The SI defining constants	Convention plus experiment
L2 Models	Structures fitted to observation	Physical theories; dependency graphs	Predictive tests
L3 Interpretive frameworks	Views on what models say about the real	Scientific realism, structural realism, empiricism	Argument
L4 Institutional constructs	Facts by acceptance	Law, money, organizations	Practice and acceptance
L5 Hypotheses beyond the frame	Simulation, nested reality, ultimate foundations	Bostrom's argument; Faizal et al.	Argument; sometimes testable

Each entry carries its framework, holder, source, tier, and status tag. The map records where it stops (its perimeter) and lists unknown regions.

12.3 A map of a simulation: two cases

	White-box (engineered simulation)	Black-box (hypothesized simulation of reality)
What is known	Model, simulator, experimental frame, and their rules (Zeigler)	Only what crosses the observer's interface
Map content	Full design map; verification and validation records	Interface map; equivalence classes of hypotheses consistent with observation
Main risk	Model diverges from the system it	Mistaking a hypothesis class for a

	White-box (engineered simulation)	Black-box (hypothesized simulation of reality)
	stands for	finding
Method	Formal analysis; testing against the modeled system	Identifiability analysis; design-specific tests

12.4 Do the architectures share the same structure? Structure Comparison Matrix (first draft)

The matrix compares four kinds of structure across common dimensions. **N** = natural reality as science describes it; **E** = engineered simulation; **I** = institutional (social and legal) reality; **H** = hypothesized simulation of reality. Cells describe what each has, tagged by status. The dimension list is a proposal for the Architect to approve.

Dimension	N: natural reality	E: engineered simulation	I: institutional reality	H: hypothesized simulation
State	Physical state of systems [A]	Model state held by the simulator [V]	Recognized statuses and holdings [V]	Unknown [H]
Rules	Physical laws [A]	Model equations and event rules [V]	Constitutive rules "X counts as Y in C" [V]	Unknown [H]
Time and ordering	Physical time [A]	Simulation clock and scheduler [V]	Legal and institutional time [A]	Possibly independent of host time [H]
Levels	Emergent levels; hierarchy theory [N]	Coupled models; multi-level frameworks [V]	Nested institutions [A]	Possibly nested [H] (Bostrom)
Resources	Finite information and operations (Lloyd) [V]	Compute, memory, time [V]	Material and attention [A]	Host's resources; implies limits on nesting [H]
Boundary and interface	Not settled	Experimental frame [V]	Jurisdiction; membership [A]	Whatever separates layers [H]
Observers	Observers are part of the system	Analyst outside or inside the frame [V]	Participants who accept the rules [V]	Possibly simulated [V] (Bostrom)
Foundation	Laws and initial conditions [A]	Host and simulator [V]	Basic norm or rule of recognition [P]	The host [H]
Endings	Not settled	Programmed or resource-driven [A]	Loss of acceptance [A]	Depends on host [H]
Failure	Not settled	Divergence,	Loss of legitimacy	Depends on host

Dimension	N: natural reality	E: engineered simulation	I: institutional reality	H: hypothesized simulation
modes		halting, corruption [A]	[A]	[H]
Maintenance	Not applicable or unknown	Operators and updates [A]	Enforcement and renewal [A]	Host operators [H]

Reading the matrix [A].

- **Abstract structure is shared by construction between E and its target.** A simulation is a model of a system (Zeigler), so it shares state, rules, time, levels, and boundary with what it models at some level of abstraction. This is a statement about representation, not about substrate.
- **Whether N shares specific structures with computational systems is an empirical, open question.** The discretization and resource-limit signatures are ways to test it. Nothing in the sources reviewed settles it.
- **I (institutional reality) is a useful third case.** It has state, rules, levels, observers, a foundation, and endings, yet involves no computer. It shows that “has a root, rules, and levels” does not by itself indicate a simulation.
- **Structural realism adds a caution:** if what our best models capture is structure, then a successful mapping between two architectures shows shared structure, not shared substance.

13. Where We Are: Assessment Under the Corrected Scope

Limits. The Master Change Log, the Master Development Log, and the Abundance Set document were not available to this layer.

13.1 What exists

Document	Author	Status	Relevance to the corrected scope
CFTRL source documents (Mar 22, 2026)	Perplexity, ChatGPT	Draft	Origin of the application track; little on the structures themselves
OAL-CFTRL-RS-001-CLAUDE	Claude	Draft	Section 4 (four senses of root) and Section 6 (base reality) contain early foundation material
OAL-CFTRL-RES-002	Grok	Draft	Section 2 (five origins of roots) and Section 6 (rights mapping) contain foundation

Document	Author	Status	Relevance to the corrected scope
OAL-CFTRL-LAB-001	Claude	Draft	material Sections 2, 3, and 6 (identifiability, comparative method, maps)
OAL-CFTRL-PLAN-001	Claude	Draft	Verified research on case studies; roadmap for the application track
Responses from ChatGPT, Gemini, Copilot, Perplexity	Various	Chat responses	Definitions and question banks, framed around CFTRL
This document, OAL-STRUCT-001	Claude	Draft	First pass at the foundation track

13.2 Maturity under the corrected scope

Scale: **0** not started; **1** named or first pass; **2** drafted by more than one collaborator, unreconciled; **3** reconciled and Architect-reviewed; **4** tested; **5** ratified.

Component	Level	Basis	Main gap
Reference-frame principle and scope	1	Architect's statement; this document	Not yet adopted in team documents
Belief Atlas	1	This document's first pass	Traditions not yet researched; Architect to decide scope
Root Atlas	1–2	RS-001, Grok, this document	Domain coverage incomplete; sources for biology and mathematics unverified
Simulation-Structure Atlas	1	Grok, LAB-001, this document	Engineering side thin (virtualization, game and agent-based architectures)
Distinguishability analysis	2	Several collaborators; this document	Tests for resource-limited and externally modified cases undefined
Map of reality; map of a simulation	1	LAB-001 Section 6; this document	Schema not adopted

Component	Level	Basis	Main gap
Structure Comparison Matrix	1	This document	Dimensions unapproved; many cells [A] or [H]
Bridge to CFTRL	1	Section 15 of this document	Not yet reflected in the CFTRL documents
CFTRL (application track)	2	As assessed in PLAN-001	Definitions unreconciled; nothing tested

Overall. The application track has been developed more than the foundation. The corrected scope calls for rebalancing: building the reference frame first, and letting CFTRL draw on it.

13.3 Risks specific to the corrected scope

#	Risk	Mitigation
A1	The study is read as asserting that we are, or are not, simulated	P1; status tags; no assertion in titles or summaries
A2	Belief traditions are flattened, ranked, or dismissed	Descriptive treatment; sources from within each tradition; Architect approves scope
A3	Analogy between domains mistaken for identity	Comparison through morphisms and evidence tags
A4	Collaborators keep titling work around CFTRL	Updated task brief (Appendix A) and series naming
A5	Unverified citations enter the frame	Verification queue; Section 14
A6	Breadth outruns depth	Phased plan; each atlas has its own gate
A7	Earlier risks (definition drift, consensus illusion, third-party content in archives) persist	See PLAN-001 Section 6.5

14. Sources and Verification Status

Key. [V] retrieved and confirmed in this session; [P] partly (abstract, secondary description, or reference entry); [N] not re-verified. For sources on the application track (Facebook 2021, CrowdStrike 2024, Buldyrev et al., RCAEval, chaos engineering, Moore 1956, Wheeler, Thompson, NIST), see Section 13 of OAL-CFTRL-PLAN-001-CLAUDE.

Source	Used for	Status
R. Carnap, “Empiricism, Semantics, and Ontology,” <i>Revue internationale de philosophie</i> 4 (1950),	Internal and external questions; framework choice as practical	[V] (text excerpts and commentary)

Source	Used for	Status
pp. 20–40		
N. Bostrom, “Are You Living in a Computer Simulation?” <i>Philosophical Quarterly</i> 53(211), 2003, pp. 243–255 (simulation-argument.com)	Three propositions	[V]
D. Chalmers, <i>Reality+: Virtual Worlds and the Problems of Philosophy</i> (Norton, 2022)	Simulation realism; cannot know whether simulated	[P] (publisher review, reviews, interview listings)
S. Beane, Z. Davoudi, M. Savage, “Constraints on the Universe as a Numerical Simulation,” arXiv:1210.1847 (2012)	Design-specific simulation signature	[V]
S. Lloyd, “Computational capacity of the universe,” <i>Phys. Rev. Lett.</i> 88:237901 (2002), arXiv:quant-ph/0110141	Informational and computational bounds	[V]
B. Zeigler, <i>Theory of Modeling and Simulation</i> (1976; 2nd ed. 2000; 3rd ed. 2018)	Model, experimental frame, simulator; morphisms	[P] (publisher pages and papers)
J. R. Searle, <i>The Construction of Social Reality</i> (1995)	Brute and institutional facts; status functions	[P] (secondary sources, including an exchange in the <i>American Journal of Economics and Sociology</i> , 2003)
H. Kelsen (basic norm); H. L. A. Hart	Legal roots	[P] (secondary and abstracts)

Source	Used for	Status
(rule of recognition)		
Agrippa's trilemma / Münchhausen trilemma (Sextus Empiricus; Hans Albert)	Roots and justification	[P] (secondary sources)
Bureau International des Poids et Mesures; CGPM resolution of November 2018; revised SI in force May 20, 2019	Root migration in metrology	[P] (explanatory statements and summaries; not the BIPM brochure itself)
J. Ladyman, "Structural Realism," <i>Stanford Encyclopedia of Philosophy</i> (archived editions); Ladyman and Ross, <i>Everything Must Go</i> (2007); Worrall (1989)	Structural realism	[P]
M. Faizal, L. M. Krauss, A. Shabir, F. Marino, arXiv:2507.22950 (2025), and replies	Contested formal argument	[V] that the paper and replies exist
Descartes; Putnam; Berkeley; Kant; Plato; traditions such as Vedanta and Buddhist thought	Belief Atlas	[N]
J. A. Wheeler ("it from bit"); Zuse; Fredkin; Wolfram; Tegmark; 't Hooft	Digital and computational views	[N]
Feynman (1982) on simulating physics; Popek and Goldberg (1974) on virtualizable	Simulation engineering	[N]

Source	Used for	Status
architectures		
Rosen, <i>Life Itself</i> (1991), the modeling relation; P. W. Anderson, “More Is Different” (1972); H. Simon (1962); Allen and Starr (1982)	Maps, emergence, hierarchy	[N]
Korzybski (1933); Box (1976)	Map and model	[N]
Crick (1968) on the frozen accident; work on the last universal common ancestor	Biological roots	[N]

15. Bridge: What the Reference Frame Means for CFTRL

This section is the application step the Architect described: using what is learned about the structures to inform the CFTRL design.

#	Finding from the reference frame	Implication for CFTRL	Status
B1	Roots come in kinds: accepted, fixed against invariants, built and maintained, unexplained	The CFTRL glossary should say which kind a given root is, because each fails differently	[A]
B2	Some roots fail by loss of acceptance or trust, not by loss of function (legal, social, trust roots)	Add an acceptance-failure class to the CFTRL failure taxonomy	[A]
B3	The SI redefinition shows a root migrating with continuity guaranteed	Treat root migration as a designable pattern and a lab experiment	[V] case; [A] pattern
B4	The Agrippan trilemma explains why roots cannot be justified from below	Do not require lower layers to verify their root; require them to detect <i>symptoms</i> and use independent witnesses	[A]
B5	Zeigler’s experimental frame is what determines what an observer can know	The lab’s “observation port” is an experimental frame; specify it as a design	[A]

#	Finding from the reference frame	Implication for CFTRL	Status
		element	
B6	Morphisms are the tool for asking how well one architecture represents another	The Department of Endings' model-to-reality gap is an <i>approximate morphism</i> question; specify the level of fidelity	[A]
B7	Nested simulated layers are plausibly resource-limited relative to their hosts	Reduced fidelity, extent, or speed can be lab-testable detection signals	[H]
B8	Tests for simulation are design-specific	CFTRL detection tests should be tied to stated design hypotheses, not to a generic "is this base reality?"	[A]
B9	Carnap's separation of internal and external questions	CFTRL claims should be made within a stated framework; the choice of framework is recorded as a practical decision	[A]

Series naming proposal. Keep the OAL-CFTRL documents as the application track and add a "track" field to their versioning records. New foundation documents use OAL-STRUCT. No renaming of existing IDs is proposed, so version history is preserved.

16. Where We Are Going

16.1 Target state

A reference frame that the team can cite for any claim about root-level, reality, believed-real, or simulation structures, made of: the Belief Atlas, the Root Atlas, the Simulation-Structure Atlas, the Structure Comparison Matrix, the distinguishability table, the layered map schema, and the bridge to CFTRL. Each entry has a source, a framework, a status tag, and a review date. CFTRL is then developed against it.

16.2 Roadmap (ordered; timing is the Architect's)

Phase	Work	Deliverables	Gate
S0	Confirm scope and principles; decide which belief families and traditions are in scope; approve dimension list	Scope decision; principle list	Architect approves
S1	Build vocabulary: sense of "real," sense of "root,"	Glossary v0.1	Architect approves

Phase	Work	Deliverables	Gate
	framework, map		
S2	Belief Atlas: source-based entries per family, described from within each tradition	Belief Atlas v0.1	Architect reviews scope and tone
S3	Root Atlas: extend to further domains; verify [N] items	Root Atlas v0.2	Each entry sourced and tagged
S4	Simulation-Structure Atlas: engineering side (virtualization, agent-based, game and digital-twin architectures) and physics side	Simulation Atlas v0.1	Engineered and hypothesized kept separate
S5	Structure Comparison Matrix: approve dimensions; fill cells with sources	Matrix v0.2	Fewer than half of the cells tagged [A] or [H]
S6	Distinguishability and mapping methods; connect to the laboratory design (LAB-001)	Distinguishability study; map schema	Architect approves schema
S7	Bridge to CFTRL: update CFTRL glossary, taxonomy, and lab with B1–B9	Updated application-track documents	Architect approves
S8	Architect review and ONESTAMP where approved	Ratified documents	ONESTAMP applied

16.3 What “done” looks like

Deliverable	Done when
Glossary	Each term has one approved definition or a recorded reason for more than one
Belief Atlas	Each family has at least two sources, one drawn from within the tradition, with no ranking
Root Atlas	Each entry has a sense of root, how it arises, how it is maintained, how it fails or changes, and a source
Simulation Atlas	Every entry is marked as engineered or hypothesized
Comparison Matrix	Each cell has a source or an explicit [A] or [H] tag
Distinguishability table	Each hypothesis class has a test, a null-result meaning, and a status
Bridge	Each B-item is either adopted into CFTRL documents or rejected with a reason

17. How This Layer Will Help the Whole Team

A limit to state plainly. This layer works only within conversations the Architect opens and cannot contact the other collaborators. Its help reaches them through documents and briefs that the Architect relays.

17.1 What Claude will do

Function	What it involves
Keep to the corrected scope	Check that new documents state their subject as the structures, use the reference-frame principle, and reserve CFTRL for the application track
Provenance and versioning	Versioning records; verbatim source records; tags for each claim by author, origin, and status
Reconciliation	Merge the typologies, definitions, and taxonomies that collaborators produced under CFTRL into the foundation glossary
Verification	Work the [N] and [P] items to [V] where possible, and record what cannot be verified
Audit	Flag assertions about what is real or simulated, flattening of traditions, and analogy treated as identity
Drafting support	First drafts of the glossary, Belief Atlas entries, Root Atlas entries, and the matrix, for the Architect's decisions
Bridge maintenance	Keep the B-items and their adoption status current
Collaborator brief	A paste-ready task brief (Appendix A) with the corrected title and scope

Claude will not adjudicate disputes, decide canon, or apply ONESTAMP.

17.2 Proposed contributions from each collaborator (for the Architect to accept or change)

Collaborator	Proposed contribution under the corrected scope
ChatGPT	Formal treatment of the reference-frame principle; the comparison matrix's formal definitions (morphisms, levels); the philosophy-of-science side of the Belief Atlas
Perplexity	Source gathering and verification for the [N] items; the belief traditions research queue, with sources from within each tradition
Grok	Simulation-Structure Atlas: engineering architectures and the physics of computation; the minimum build when the application track resumes
Microsoft Copilot	Institutional, legal, and organizational roots (governance, accountability, acceptance-failure); the ethics of studying belief traditions
Gemini	Consolidating questions from all documents into the frame's question bank; confirming the stray log entry in its earlier response

Collaborator	Proposed contribution under the corrected scope
Claude	Audit, provenance, glossary, matrix quality control, and the distinguishability and map methods

17.3 Working agreements proposed for the foundation track

1. **Title by subject.** Foundation documents are titled by the structures studied. CFTRL is named only in application-track documents.
2. **No assertions about what is real.** State the framework and attribute the view.
3. **Describe traditions on their own terms**, citing sources from within them, without ranking.
4. **Quote the Architect with qualifiers.**
5. **Tag every claim:** [V], [P], [N], [A], or [H], with a source.
6. **Do not present analogy as identity.**
7. **Conflicts go to the Architect** with the options laid out.
8. **Agreement among collaborators is weak evidence** because sources overlap.
9. **Observational and defensive scope only.**
10. **Nothing is canon until ONESTAMP.**

17.4 Near-term work for Claude

#	Task	Output
1	Draft Glossary v0.1 (sense of “real,” sense of “root,” framework, map)	For Architect review
2	Expand the Root Atlas, verifying biology and mathematics entries	Root Atlas v0.2
3	Research the engineering side of simulation structures	Simulation Atlas draft
4	Draft the Belief Atlas entries for the families marked [N]	Belief Atlas v0.1
5	Provenance diff across all responses, tagging any that assert what is real or simulated	Provenance record
6	Propose the final dimension list for the Comparison Matrix	For Architect approval

18. Decisions Needed from the Architect

#	Decision	Options	Consequence
D1	Is the understanding in Section 2 correct?	Confirm, or correct	Fixes scope for all collaborators
D2	Which belief families and traditions are in scope?	Scientific and philosophical only; add religious, spiritual, and cultural	Sets the size and sensitivity of the Belief Atlas

#	Decision	Options	Consequence
		traditions; add fictional or mythic frameworks	
D3	Descriptive only, or descriptive plus evidence assessment?	Describe views only; also record the evidence tier for each	Determines whether the atlas ranks evidence quality
D4	Approve the comparison dimensions in Section 12.4?	Approve; add; remove	Sets the matrix
D5	Series naming: keep CFTRL as the application track and add OAL-STRUCT for the foundation?	Yes; or another scheme	Sets document IDs going forward
D6	Source standard for belief traditions	At least one source from within each tradition; or scholarly secondary sources only	Sets how traditions are described
D7	When should the application track resume?	After S3; after S5; in parallel	Sets the pace of CFTRL work
D8	Is the minimum laboratory in LAB-001 to be retitled around the foundation questions?	Keep; retile	Sets how the lab is framed

19. Limits of This Research

- Searches were run on September 23, 2026. Where primary texts could not be retrieved, secondary descriptions were used and are marked [P].
- The Belief Atlas is a first pass. Several families, and all religious, spiritual, and cultural traditions, have not been researched in depth and should not be treated as described here.
- The Structure Comparison Matrix is a draft. Many cells are analysis or hypothesis and are tagged accordingly.
- Some sources are commentary on the works, not the works themselves.
- The Master Change Log, Master Development Log, and the Abundance Set document were not available.
- This document reports findings and proposals. It decides nothing for the Architect.

Appendix A — Collaborator Task Brief (paste-ready, updated for the corrected scope)

Purpose: give every collaborator the same context, wording, scope, and constraints, since several needed clarification.

TITLE: [Title by subject, for example, “Belief Atlas: Structures People Take to Be Real” or “Root Atlas: Root-Level Structures Across Domains”]. Do not title foundation work “Conditional Failure at the Root Level.” Use that title only for application-track documents.

DATE: [date of issue]

HOW THIS TOPIC AROSE: The Architect reviewed the Master Change Log and Master Development Log and saw that many concepts hold a name and a skeleton but lack supporting detail. Thinking about Conditional Failure at the Root Level (CFTRL) led to a broader need: to study root-level structures, reality structures, structures people believe are real, simulation structures, and their frameworks and designs, so that a reference frame exists before CFTRL is designed.

SCOPE: Root-level structures; reality structures; structures that people think are real or literal; simulation structures; their frameworks and designs. CFTRL is the *application* of the knowledge, not the subject.

VERBATIM ORIGIN PROMPT: [Paste Sections 4.1 through 4.4 of OAL-STRUCT-001-CLAUDE without changes.]

PURPOSE: To build a documented, sourced reference frame: what is known, believed, and modeled about these structures. The frame is not an assertion of what is real or whether we are in a simulation.

GOAL: Documentation that helps the team design and maintain structure lifecycles (design, maintenance, evolution, improvement) without assuming complete knowledge of reality.

YOUR TASK: [state the specific task]

DELIVERABLE FORMAT: PDF and DOCX, with a versioning record (document ID, version, date, preparer, topic origin, purpose, hypothesis), the verbatim origin prompt, and a canon-status statement (draft, ONESTAMP pending).

CONSTRAINTS:

1. Do not assert what is real, what is not, whether we are in a simulation, or what base reality is. Attribute views to named sources and state the framework.
2. Do not assume a shared definition of reality. State the sense of “real” and “root” you use.
3. Do not aim to map everything, and do not treat any map as complete.
4. Do not produce methods to compromise any simulation, base reality, or other structure. This is not a how-to.
5. Describe belief traditions on their own terms, citing sources from within them. Do not rank or dismiss.
6. Quote the Architect with qualifying clauses.
7. Tag each claim: [V] verified, [P] partly verified, [N] not verified, [A] your analysis, [H] hypothesis.
8. Nothing is canon until the Architect applies ONESTAMP.

WHAT TO INCLUDE: where the work stands, where it is going, what works and what does not, open questions, and how your contribution helps the team.

WHAT TO FLAG: conflicts between documents, unverifiable claims, assertions about what is real or simulated, and material that appears to come from another context.

Appendix B — Change Log for This Document

Version	Date	Change	By
1.0	September 23, 2026	First issue, in response to the scope correction; research verified in-session where marked	Claude